

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407796
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Yamamoto; Yuki

Surrounding image display apparatus that displays surrounding image when driver intends to make a lane departure and method

Abstract

A surrounding image display apparatus that comprises an imaging device that captures an image of a surrounding of a vehicle, a display device that displays a surrounding image captured by the imaging device, and an electronic control unit that controls the display device, the electronic control unit being configured to start displaying the surrounding image when it is determined that a driver intends to make a lane departure, and end the displaying of the surrounding image when it is determined that the driver has performed a steering operation without visually confirming a situation of a blind spot on the side of the lane departure.

Inventors: Yamamoto; Yuki (Toyota, JP)

Applicant: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP); DENSO CORPORATION (Kariya, JP)

Family ID: 90243385

Assignee: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP); DENSO CORPORATION (Kariya, JP)

Appl. No.: 18/367732

Filed: September 13, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240098227 A1	Mar. 21, 2024

Foreign Application Priority Data

JP	2022-148136	Sep. 16, 2022
----	-------------	---------------

Publication Classification

Int. Cl.: H04N7/18 (20060101); B60R1/27 (20220101)

U.S. Cl.:

CPC H04N7/188 (20130101); B60R1/27 (20220101); B60R2300/105 (20130101); B60R2300/8026 (20130101)

Field of Classification Search

CPC: H04N (7/188); H04N (7/18); H04N (7/183); B60R (1/27); B60R (2300/105); B60R (2300/8026); B60R (1/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9969261	12/2017	Kodama	N/A	N/A
10045173	12/2017	Morimura	N/A	N/A
10106157	12/2017	Sawada	N/A	N/A
10150407	12/2017	Takahashi	N/A	N/A
10696297	12/2019	Nguyen Van	N/A	N/A
11001255	12/2020	Fukuman	N/A	N/A
11110937	12/2020	Kinoshita	N/A	N/A
2012/0242834	12/2011	Satoh	348/148	G06T 11/00
2019/0001968	12/2018	Yorifuji	N/A	N/A
2019/0344828	12/2018	Omori	N/A	N/A
2019/0389488	12/2018	Yamada	N/A	N/A
2020/0369205	12/2019	Nakai	N/A	N/A
2023/0012629	12/2022	Shin	N/A	G06V 20/588
2024/0187556	12/2023	Yamamoto	N/A	G06V 20/588

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2017147505	12/2016	JP	H04N 23/90
2019-110389	12/2018	JP	N/A
20130117430	12/2012	KR	B60W 40/02
101353541	12/2013	KR	B60R 1/00
20140033647	12/2013	KR	B60R 1/00

Primary Examiner: Pontius; James M

Attorney, Agent or Firm: Dinsmore & Shohl LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. JP2022-148136 filed on Sep. 16, 2022, the content of which is hereby incorporated by reference in its entirety into this application.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a surrounding image display apparatus and method, and a non-transitory computer-readable storage medium storing a program for executing a surrounding image display control.

2. Description of the Related Art

(3) As one of driving support devices for vehicles such as automobiles, a surrounding image display apparatus has been known that displays a surrounding image such as a rear image and a side image captured by an imaging device on a display device in a situation where a driver intends to make a lane departure, such as when changing lanes or turning left or right.

(4) For example, Japanese Patent Application Laid-open No. 2019-110389 describes a surrounding image display apparatus configured to display a surrounding image on a display device and to switch the displayed image based on a driver's operation of a turn lever, a touch sensor, or the like.

(5) According to this type of surrounding image display apparatus, a driver can look at an image displayed on a display device by directing his or her line of sight toward the display device, thereby allowing the driver to easily confirm a situation of the rear or the like of a vehicle without significantly changing a direction of a head or a posture of an upper body.

(6) Since a range that can be captured by an imaging device and the number of images that can be displayed on the display device are limited, there is a blind spot area where a situation cannot be confirmed even by looking at the image displayed on the display device. Therefore, if the driver gazes at the image displayed on the display device in a situation where the driver intends to make a lane departure, there is a risk that the driver will neglect to confirm a situation around the vehicle, especially a situation in the blind spot area.

SUMMARY

(7) The present disclosure provides a surrounding image display apparatus and method improved so as to reduce a possibility that a driver will neglect to confirm a situation of a blind spot area when the driver intends to make a lane departure.

(8) According to the present disclosure, a surrounding image display apparatus is provided that comprises an imaging device that captures an image of a surrounding of a vehicle, a display device that displays a surrounding image captured by the imaging device, and an electronic control unit that controls the display device.

(9) The electronic control unit is configured to start displaying the surrounding image when it is determined that a driver intends to make a lane departure, and end the displaying of the surrounding image when it is determined that the driver has performed a steering operation without visually confirming a situation of a blind spot on the side of the lane departure.

(10) According to the present disclosure, a surrounding image display method is provided that comprises a step of displaying a surrounding image captured by an imaging device that captures an image of a surrounding of a vehicle on a display device.

(11) The surrounding image display method further comprises a step of starting displaying the surrounding image when it is determined that a driver intends to make a lane departure, and a step of ending the displaying of the surrounding image when it is determined that the driver has performed a steering operation without visually confirming a situation of a blind spot on the side of the lane departure.

(12) Further, according to the present disclosure, a non-transitory computer-readable storage medium storing a program for executing a surrounding image display control is provided. The

program causes the computer to implement a step of displaying a surrounding image captured by an imaging device that captures an image of a surrounding of the vehicle.

(13) The program further comprises a step of starting displaying the surrounding image when it is determined that a driver intends to make a lane departure, and a step of ending the displaying of the surrounding image when it is determined that the driver has performed a steering operation without visually confirming a situation of a blind spot on the side of the lane departure.

(14) According to the above surrounding image display apparatus and method, and the non-transitory computer-readable storage medium, when it is determined that a driver intends to make a lane departure, displaying of the surrounding image is started, and when it is determined that the driver has performed a steering operation without visually confirming a situation of a blind spot area on the side of the lane departure, the displaying of the surrounding image ends. Therefore, since the driver cannot see the surrounding image, a possibility of the driver visually confirming a situation of a blind spot area on the side of the lane departure is increased, which enables to reduce the possibility that the driver will neglect to confirm the situation of the blind spot area when the driver intends to make a lane departure. Incidentally, 'visually confirming a situation' means to confirm a situation by visual observation directly or by way of a mirror without looking at a display device.

(15) In one aspect of the present disclosure, the electronic control unit is configured to determine that the driver intends to make a lane departure when it is determined that the driver has confirmed the rear of the vehicle using a rearview mirror and tilted a turn lever within a preset determination time.

(16) In general, when a driver intends to make a lane departure, the driver confirms the rear of a vehicle using a rearview mirror and tilts a turn lever as a series of actions. Therefore, it can be determined whether or not the driver intends to make a lane departure by determining whether or not the driver has confirmed the rear of the vehicle using the rearview mirror and tilted the turn lever within a preset determination time.

(17) According to the above aspect, when it is determined that a confirming of the rear by the rear view mirror and a tilting operation of the turn lever have been performed by the driver within the preset determination time, it is determined that the driver intends to make the lane departure. Therefore, when the driver intends to deviate from the lane, it is possible to determine that the driver intends to make the lane departure.

(18) In another aspect of the present disclosure, the electronic control unit is configured to end the displaying of the surrounding image when it is determined that the driver gazes at the display device for a preset reference time or more.

(19) If a driver gazes at a display device for a long period of time, the possibility of the driver visually confirming a situation of a blind spot on the side of the lane departure decreases.

(20) According to the above aspect, when it is determined that the driver gazes at the display device for a preset reference time or more, the display of the surrounding image is terminated. Therefore, since the driver can no longer see the surrounding image, it is possible to increase the possibility that the driver confirms the situation of the blind spot area on the side of the lane departure by visual observation.

(21) Other objects, other features and attendant advantages of the present disclosure will be readily understood from the description of the embodiments of the present disclosure described with reference to the following drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic configuration diagram showing a surrounding image display apparatus

according to the first embodiment.

(2) FIG. 2 is a flowchart showing a surrounding image display control routine in the first embodiment.

(3) FIG. 3 is a flowchart showing a surrounding image display control routine in the second embodiment.

(4) FIG. 4 is a diagram illustrating a line of sight of a driver when the driver intends to move a vehicle to a right lane.

(5) FIG. 5 is a diagram showing a situation in which the line of sight of the driver is directed to a display device.

(6) FIG. 6 is a diagram illustrating a situation in which a vehicle straddles a lane boundary line.

DETAILED DESCRIPTION

(7) Referring to the accompanying drawings, a surrounding image display apparatus and method according to the embodiments of the present disclosure will be described in detail.

First Embodiment

(8) As shown in FIG. 1, a surrounding image display apparatus **100** according to the first embodiment of the present disclosure is applied to a vehicle **102** and includes a driving assistance ECU **10**. The vehicle **102** may be a vehicle capable of automatic driving and includes a multimedia ECU **20** and a steering ECU **30**. The ECU is an electronic control unit comprising a microcomputer as its main part. In the following description, the vehicle **102** is referred to as an own vehicle **102** as required in order to distinguish it from other vehicles.

(9) The microcomputer in each ECU includes a CPU, a ROM, a RAM, readable and writable non-volatile memories (N/M), interfaces (I/F), and the like. The CPU realizes various functions by executing instructions (programs and routines) stored in the ROM. Furthermore, these ECUs are connected to each other so as to be data-interchangeable (communicating) via a CAN (Controller Area Network) **104**. Thus, information supplied by sensors (including switches) connected to a particular ECU may be transmitted to other ECUs.

(10) The driving assistance ECU **10** is a main electronic control unit which carries out driving assistance controls such as lane deviation prevention control and inter-vehicular distance control. In the embodiment, the driving assistance ECU **10** cooperates with other ECUs to perform surrounding image display control so as to improve security when making a lane departure, as will be more fully described below.

(11) The driving assistance ECU **10** is connected with a camera sensor **12**, a radar sensor **14** and a monitor device **16**. The camera sensor **12** functions as an imaging device that captures an image of a surrounding of the own vehicle **102**. The camera sensor **12** and the radar sensor **14** function as a target information acquiring device **13** that acquires target information around the own vehicle **102**. The camera sensor **12** includes four camera sensors that capture front, rear, right and left side areas of the vehicle, but is not limited to four. The radar sensor **14** includes five radar sensors that acquire target information of a solid object in the forward area, the right front area, the left front area, the right rear area, and the left rear area, but is not limited to five.

(12) Although not shown in the figure, each camera sensor of the camera sensor **12** includes a camera unit for capturing a surrounding of the vehicle **102** and a recognition unit for recognizing a target such as a white line of a road and another vehicle by analyzing an image data obtained by capturing by the camera unit. The recognition unit supplies information about the recognized target to the driving assistance ECU **10** at predetermined time intervals.

(13) Although not shown in the figure, each radar device of the radar sensor **14** includes a radar transceiver and a signal processing unit. The radar transceiver radiates a millimeter-wave band radio wave (hereinafter, referred to as "millimeter wave") toward the surrounding of the vehicle **102**, and receives millimeter waves (i.e., reflected waves) reflected by a solid object (e.g., another vehicle, a bicycle, etc.) present in a radiation range. The signal processor acquires information representing a relative distance and a relative velocity between the own vehicle and the solid

object, a relative position (direction) of the solid object with respect to the own vehicle, and supplies the acquired information to the driving assistance ECU **10** at predetermined time intervals based on a phase difference between the radiated millimeter wave and the received reflected wave, an attenuation level of the reflected wave, and a time from the radiation of the millimeter wave to the reception of the reflected wave. In place of or in addition to the radar sensor **14**, LiDAR (Light Detection And Ranging) may be used.

(14) The monitor device **16** includes a driver imaging camera and an identification unit, although not shown in the figure. The driver imaging camera is installed in front of a driver's seat to acquire a driver's face image by imaging a driver's face and supply it to the identification unit. The identification unit is preliminarily learned to detect positions of a pupil and a corneal reflection image of a light source, and identifies positions of the pupil and the corneal reflection image in a driver's eye included in the face image. The monitor device **16** detects a direction of the driver's line of sight based on a positional relationship between the pupil and the corneal reflection image.

(15) The identification unit may be, for example, a convolutional neural network (CNN) having a plurality of layers connected in series from an input side to an output side. Face images including pupils and corneal reflection images are inputted to the CNN as teacher data, and training is performed, so that the CNN operates as a discriminator to identify positions of a pupil and a corneal reflection image. The monitor device **16** that detects a direction of the driver's line of sight may be a device of any configuration known in the art.

(16) Furthermore, the driving assistance ECU **10** is connected with a display switch **18**. The display switch **18** is provided in a position to be operated by the driver. As will be described in more detail below, when the display switch **18** is turned on, the driving assistance ECU **10** starts capturing an image of the rear of the vehicle by the camera sensor **12** and displaying the captured image.

(17) The multimedia ECU **20** is connected with a display device **22** and a microphone **24**. The display device **22** may be, for example, a head-up display or a multi-information display on which meters and various types of information are displayed, and may be a display of a navigation device.

(18) The microphone **24** converts voice into an electrical signal to provide it to the multimedia ECU **20**. The multimedia ECU **20**, when it is determined by analyzing the electric signal that a command to display the image of the rear of the vehicle is included in the voice, outputs a command signal of the image display to the driving assistance ECU **10**. Therefore, the driving assistance ECU **10**, even when the display switch **18** is off, starts capturing an image of the rear of the vehicle by the camera sensor **12** and displaying the image.

(19) The steering ECU **30** is connected with a turn lever **32** and turn lamps **34R**, **34L**. When the turn lever **32** is tilted in the vertical direction corresponding to the right-turn direction and the left-turn direction, it supplies a signal indicating the tilting to the steering ECU **30** to blink the right and left turn lamps **34R** and **34L**, respectively.

(20) Although not shown in FIG. **1**, the CAN **104** is connected with a drive ECU for controlling a drive device such as an engine and a brake ECU for controlling a brake device. The drive ECU controls a driving force of the vehicle by controlling the drive device, and the brake ECU controls a braking force of the vehicle by controlling the brake device.

(21) Furthermore, the CAN **104** is connected with a driving operation sensor **40** and a vehicle status sensor **50**. Information detected by the driving operation sensor **40** and the vehicle status sensor **50** (referred to as sensor information) is transmitted to the CAN **104**. The sensor information transmitted to the CAN **104** is available in the respective ECUs as appropriate. Incidentally, the sensor information may be information of a sensor connected to a particular ECU and may be transmitted from that particular ECU to the CAN **104**.

(22) The driving operation sensor **40** includes a driving operation amount sensor that detects an operation amount on an accelerator pedal, a braking operation amount sensor that detects a pressing force on a brake pedal, a master cylinder pressure and the like. Furthermore, the driving operation sensor **40** includes a steering angle sensor that detects a steering angle θ , a steering torque sensor

that detects a steering torque T_s , and the like.

(23) The vehicle status sensor **50** includes a vehicle speed sensor that detects a vehicle speed of the vehicle **102**, a longitudinal acceleration sensor that detects a longitudinal acceleration of the vehicle, a lateral acceleration sensor that detects a lateral acceleration of the vehicle, and a yaw rate sensor that detects a yaw rate of the vehicle.

(24) The ROM of the driving assistance ECU **10** stores a program of the surrounding image display control corresponding to the flowchart shown in FIG. 2. The CPU of the driving assistance ECU **10** reads the control program from the ROM to the RAM and executes the surrounding image display control in accordance with the flowchart shown in FIG. 2, as will be described in more detail later.

(25) <Surrounding Image Display Control Program in the First Embodiment>

(26) Next, the surrounding image display control program in the first embodiment will be described with reference to the flowchart shown in FIG. 2. The control in accordance with the flowchart shown in FIG. 2 is repeatedly performed by the CPU at predetermined time intervals when an ignition switch, not shown in FIG. 1, is on. In the first embodiment, the surrounding image display method in the first embodiment is executed by executing the surrounding image display control according to the flowchart shown in FIG. 2.

(27) First, in step **S10**, the CPU determines whether or not a driver intends to make a lane departure by determining whether or not the driver has performed rear-view confirmation by a rearview mirror and tilting operation of the turn lever **32** within a preset determination time $\Delta t1$ (a positive constant). When a negative determination is made, the CPU advances the present control to the step **S30**, and when an affirmative determination is made, the CPU advances the present control to step **S20**.

(28) Incidentally, the determination of whether or not the driver has performed the rear-view confirmation by the rearview mirror is performed based on a direction of a driver's line of sight detected by the monitor device **16**. Further, the determination of whether or not the driver has performed the tilting operation of the turn lever **32** is performed based on the information about the turn lever supplied from the steering ECU **30**. Further, in some embodiments, the rearview mirror may be a room mirror, but in some embodiments the rearview mirror is a side mirror such as a door mirror or a fender mirror.

(29) An affirmative determination is made in step **S10** in either of the following cases. Incidentally, when the side of the turn lamp which is flashed by the tilting of the turn lever and the side of the direction of the driver's line of sight with respect to the front-rear direction of the vehicle are different from each other, for example, when the former is left and the latter is right, an affirmative determination is not made in any of the following cases. **X1**: When it is determined that the driver has performed the tilting operation of the turn lever within a preset determination time $\Delta t1$ after it is determined that the driver performed the rear-view confirmation by the rearview mirror. **X2**: When it is determined that the driver has performed the rear-view confirmation by the rearview mirror within the preset determination time $\Delta t1$ after it is determined that the driver performed the tilting operation of the turn lever.

(30) For example, FIG. 4 illustrates a case in which a driver **60** attempts to move the vehicle **102** to the right lane. As shown in the figure, when it is detected that a line of sight **62** of the driver **60** is directed to a right door mirror **64** by the monitor device **16**, it may be determined that the driver has performed the rear-view confirmation by the rear-view mirror.

(31) In step **S20**, the CPU calculates an amount of change $\Delta\theta$ of the steering angle θ within a preset determination time $\Delta t2$ (a positive constant), and determines whether or not an absolute value of the amount of change $\Delta\theta$ is equal to or greater than a reference amount of change $\Delta\theta_c$ (a positive constant), that is, whether or not a steering operation is performed by the driver. When an affirmative determination is made, the CPU terminates the present control once, and when a negative determination is made, the CPU advances the present control to step **S40**. Incidentally, step **S20** is performed in order not to show the surrounding image in a situation where the steering

operation is being performed by the driver.

(32) In step **S30**, the CPU determines whether or not the display switch **18** is on, or whether or not a voice instruction of the surrounding image display is inputted via the microphone **24** based on the information supplied from the multimedia ECU **20**. When the CPU determines that the display switch **18** is off and that no voice instruction of the surrounding image display is inputted, a negative determination is made and the CPU terminates the present control once. In contrast, when the CPU determines that the display switch **18** is on, or when the CPU determines that the voice instruction of the surrounding image display is inputted, an affirmative determination is made and the CPU advances the present control to step **S40**. Therefore, in step **S30**, the CPU determines whether or not the driver wants the displaying of the surrounding image.

(33) In step **S40**, the CPU starts capturing an image of the rear of the vehicle **102** and the side of the vehicle on the lane departure side by the camera sensor **12**, starts displaying the surrounding image acquired by the capturing, and sets a reference steering angle θ_r used in a determination of step **S80** to be described later to a present steering angle θ .

(34) In step **S50**, the CPU determines whether or not the turn lever **32** has been returned to a neutral position by the driver based on the information about the turn lever supplied from the steering ECU **30**. When an affirmative determination is made, the CPU advances the present control to step **S100**, and when a negative determination is made, the CPU advances the present control to step **S60**.

(35) In step **S60**, the CPU determines whether or not the display switch **18** is off or whether or not a voice end command of the surrounding image display is inputted via the microphone **24** based on the information supplied from the multimedia ECU **20**. When it is determined that the display switch **18** is off, or when it is determined that the voice end command of the surrounding image display is inputted, an affirmative determination is made, and the CPU advances the present control to step **S100**. In contrast, when it is determined that the display switch **18** is on, or it is determined that the voice end command of the surrounding image display is not inputted, a negative determination is made, and the CPU advances the present control to step **S70**.

(36) In step **S70**, the CPU determines whether or not the driver gazes at a surrounding image displayed on the display device **22** for a reference period of time (a positive constant) or more based on the line of sight of the driver detected by the monitor device **16**. When an affirmative determination is made, the CPU advances the present control to step **S100**, and when a negative determination is made, the CPU advances the present control to step **S80**. Incidentally, as shown in FIG. 5, when it is detected by the monitor device **16** that the line of sight **62** of the driver **60** is directed to the display device **22** for a period of time that is equal to or greater than the reference period of time, it may be determined that the driver gazes at a surrounding image displayed on the display device **22** for the a reference period of time or more.

(37) In step **S80**, the CPU determines whether or not an absolute value of a difference $\theta - \theta_r$ between a present steering angle θ and the reference steering angle θ_r set in step **S40** is equal to or greater than a reference value $\Delta\theta_r$ (a positive constant), that is, whether or not a steering operation has been performed by the driver. When a negative determination is made, the CPU returns the present control to step **S50**, and when an affirmative determination is made, the CPU advances the present control to step **S90**.

(38) In step **S90**, the CPU determines whether or not the driver has confirmed a state of a blind spot area. When an affirmative determination is made, the CPU returns the present control to step **S50**, and when a negative determination is made, the CPU advances the present control to step **S100**. In this case, when it is determined based on the direction of the driver's line of sight detected by the monitor device **16** that the driver's line of sight is directed to a side mirror such as a door mirror or a fender mirror on the side where the turn lever **32** is tilted, an affirmative determination may be made. Incidentally, in FIG. 4, the hatched area **66** shows an example of a blind spot area on which a situation confirmation can be made using a door mirror **64** on the right side.

(39) Thus, in steps **S80** and **S90**, it is determined whether or not the driver performed a steering

operation without visually confirming the state of the blind spot area on the side of the lane departure. When affirmative and negative determinations are made in steps **S80** and **S90**, respectively, it is determined that the driver performed a steering operation without visually confirming the state of the blind spot area on the side of the lane departure.

(40) In step **S100**, the CPU terminates capturing an image of the rear of the vehicle **102** by the camera sensor **12**, terminates the display of the surrounding image acquired by the capturing and clears the reference steering angle θ_r .

(41) As will be understood from the above description, according to the first embodiment, in step **S10**, it is determined that the driver intends to make a lane departure, and when an affirmative determination is made, the displaying of a surrounding image is started in step **S40**.

(42) Further, when it is determined in steps **S80** and **S90** that the driver performed a steering operation without visually confirming the state of the blind spot area on the side of the lane departure, in step **S100**, the displaying of the surrounding image is terminated.

Second Embodiment

(43) The surrounding image display apparatus according to the second embodiment also has the same configuration as the configuration shown in FIG. 1. The ROM of the driving assistance ECU **10** stores the surrounding image display control program in the second embodiment corresponding to the flowchart shown in FIG. 3.

(44) The surrounding image display control by the flowchart shown in FIG. 3 is also repeatedly performed by the CPU of the driving assistance ECU **10** when the ignition switch is on. In the second embodiment, the surrounding image display method in the second embodiment is executed by executing the surrounding image display control according to the flowchart shown in FIG. 3.

(45) As can be seen from the comparison of FIGS. 3 and 2, the CPU of the driving assistance ECU **10** performs steps **S25** and **S85** instead of steps **S20** and **S80**, respectively, in the first embodiment. In steps **S25** and **S85**, the CPU determines whether or not the vehicle **102** is straddling a lane border line **68** on the side of the lane departure, as shown in FIG. 6, based on information provided by the camera sensor **12** that captures an image of the front of the vehicle **102**, for example. It should be noted that when it is determined that a portion of the vehicle **102** is in a lane **70** ahead of the lane departure and the remaining portion of the vehicle **102** is in an original lane **72**, it may be determined that the vehicle is straddling the lane border line **68** on the side of the lane departure.

(46) As can be seen from the above description, the surrounding image display apparatus according to the second embodiment operates in the same manner as the surrounding image display apparatus according to the first embodiment except that it is determined whether or not the driver is steering in a lane departure direction by determining whether or not the vehicle **102** is straddling a lane border line **68** on the side of a lane departure.

(47) Therefore, also in the second embodiment, in step **S10**, it is determined whether or not the driver intends to make a lane departure, and when an affirmative determination is made, the displaying of the surrounding image is started in step **S40**.

(48) Further, when it is determined in steps **S85** and **S90** that the driver performed a steering operation without confirming visually a state of the blind spot area on the side where a lane departure is made, in step **S100**, the displaying of the surrounding images is terminated.

(49) Therefore, according to the first and second embodiments, since the driver cannot see a surrounding image due to the end of the display of the surrounding image, it is possible to increase the possibility that the driver will visually confirm a situation of the blind spot area on the side of the lane departure, thereby reducing the risk of the driver neglecting to confirm the situation of the blind spot area.

(50) In particular, according to the first and second embodiments, when it is determined that a confirming of the rear by the rear view mirror and a tilting operation of the turn lever have been performed by the driver within the preset determination time $\Delta t1$ (**S10**), it is determined that the driver intends to make a lane departure. Therefore, when the driver intends to make the vehicle

depart from a lane, it is possible to determine that the driver intends to make a lane departure.

(51) Further, according to the first and second embodiments, when it is determined that the driver gazes at the display device for a preset reference time or more (S70), the displaying of the surrounding image is terminated. Therefore, since the driver can no longer see the surrounding image, it is possible to increase the possibility that the driver visually confirms the situation of the blind spot area on the side of the lane departure.

(52) Further, according to the first and second embodiments, even if it is determined that the driver intends to make a lane departure, when it is determined that a steering operation is performed by the driver (S20, S25), the displaying of the surrounding image is not started. Therefore, since the surrounding image is not displayed in the situation where a steering operation is performed by the driver, it is possible to prevent the driver from moving his or her line of sight to the display device during the steering operation. Therefore, when it is determined that the driver intends to make a lane departure, it is possible to improve safety in a situation in which the driver tries to make the vehicle depart from the lane as compared to where a surrounding image is displayed regardless of whether or not a steering operation is performed by the driver.

(53) While the present disclosure has been described in detail with reference to specific embodiments, it will be apparent to those skilled in the art that the present disclosure is not limited to the embodiments described above, and various other embodiments are possible within the scope of the present disclosure.

(54) For example, in the first and second embodiments described above, in step S40, capturing of an image and displaying of a surrounding image are started, in step S100, the capturing of an image and displaying of a surrounding image are terminated. However, image capturing by the camera sensor 12 may always be performed, in step S40, the displaying of the surrounding image may be started, and in step S100, the displaying of the surrounding image may be terminated.

(55) In the above-described first embodiment, in step S10, it is determined whether or not a driver intends to make a lane departure, and in step S20, it is determined whether or not a steering operation is performed by the driver. Similarly, in the above-described second embodiment, in step S10, it is determined whether or not a driver intends to make a lane departure, and in step S25, it is determined whether or not a steering operation is performed by the driver. However, steps S20 and S25 may be omitted.

(56) Further, the above-described first and second embodiments may be integrated. For example, when a negative determination is made in step S20, step S25 may be performed. Further, when a negative determination is made in step S80, step S85 may be performed.

Claims

1. A surrounding image display apparatus comprising: a camera that captures an image of a surrounding of a vehicle; a display that displays a surrounding image captured by the camera; and an electronic control unit that controls the display, the electronic control unit is configured to start displaying the surrounding image when it is determined that a driver intends to make a lane departure, and end the displaying of the surrounding image when it is determined that the driver has performed a steering operation without visually confirming a situation of a blind spot on the side of the lane departure.
2. The surrounding image display apparatus according to claim 1, wherein the electronic control unit is configured to determine that the driver intends to make a lane departure when it is determined that the driver has confirmed the rear of the vehicle using a rearview mirror and tilted a turn lever within a preset determination time.
3. The surrounding image display apparatus according to claim 1, wherein the electronic control unit is configured to end the displaying of the surrounding image when it is determined that the driver gazes at the display for a preset reference time or more.

4. A surrounding image display method comprising: displaying a surrounding image captured by camera that captures an image of a surrounding of a vehicle on a display; starting displaying the surrounding image when it is determined that a driver intends to make a lane departure; and ending the displaying of the surrounding image when it is determined that the driver has performed a steering operation without visually confirming a situation of a blind spot on the side of the lane departure.
 5. A non-transitory computer-readable storage medium storing a program for executing a surrounding image display control, the program causing a computer to: display a surrounding image captured by a camera that captures an image of a surrounding of the vehicle; start displaying the surrounding image when it is determined that a driver intends to make a lane departure; and end the displaying of the surrounding image when it is determined that the driver has performed a steering operation without visually confirming a situation of a blind spot on the side of the lane departure.
-